LIGHTYEAR

MS 108 CardDemo PostgreSQL data migration proof

Recorded October 15 2026. Implementation: PRs #350 and #351. This record extends the mainframe data foundations in MS18 and MS19.

LIGHTYEAR can derive a bounded PostgreSQL data model from CardDemo's public mainframe definitions, load its public records, and check whether the stored values reconstruct the original bytes. This gives a customer a concrete way to find data loss, changed decimal values and incorrect key ordering before a migration expands to a real estate. The overall migration verdict remains **indeterminate** because important IMS and application behavior is still unverified.

From source definitions to measured data

The source inventory pins 24 public upstream files to commit `59cc6c2fd7ebd7ef7925cad552a01a4b8b6e4d5e`. Copybooks describe field layouts, IDCAMS definitions describe VSAM files and keys, and DBD definitions describe IMS segments. The generator checks their hashes and computes mappings, PostgreSQL DDL, primary keys, alternate indexes and child tables for supported repeated fields.

The loader decodes EBCDIC text, packed and binary numbers, implied decimals and supported OCCURS layouts. It preserves fixed-width values, filler bytes and encoding details needed for exact reconstruction. Invalid records go to quarantine with their original bytes and reason. The proof reads typed values back from PostgreSQL and reconstructs records; returning a saved copy of each complete input record does not satisfy the round-trip check.

Measured PostgreSQL results

The retained measurement comes from successful CI run 38816713103 at implementation commit `aad91e37a4b84d0d92c9cd886bb44d5a0915c036`, using PostgreSQL 16. The final original PR head also passed the dedicated data workflow. Database workloads ran in CI only.

Dataset	Loaded and reconstructed	Browse comparisons
Account	50 of 50	212 of 212
Customer	50 of 50	212 of 212
Card	50 of 50	540 of 540
Card cross-reference	50 of 50	540 of 540
User security	10 of 10	56 of 56
Total	210 of 210	1560 of 1560

All five dataset reconciliations passed, with zero quarantined records in this public sample. Browse checks cover observed starting keys, prefixes, forward and reverse traversal, low and high boundaries, and nonunique alternate-key ordering. An independent byte-order calculation supplies the expected order.

A generated IMS fixture contains two root segments and four child segments. All six reconstructed exactly and the single generated traversal stream matched. This measures the generated fixture; it does not validate the retained public IMS utility unload or a full DL/I implementation.

Boundaries and next evidence

Public IMS unload framing and unresolved logical relationships still need verification. Full stateful DL/I replay and the Java access API remain open. REDEFINES is refused without a reviewed discriminator and live-reader policy; nested repeated-field migration is also refused. Unsupported declarations cannot silently become successful migrations.

Screens and CICS web journeys were outside this data delivery: zero screen sessions were compared and screen mutation detection was not measured. The supplied Compose bundle is data-only and was not run locally. There is no customer workload or z/OS equivalence claim and no release authorization.

The Factory review request is unsigned and covers only the measured data gates. The published JSON explicitly identifies itself as a derived summary. The original 1,230,556-byte receipt remains in the CI artifact and local staging pending the designated retention store; the summary preserves its hash without pretending to be the complete receipt.

Supporting records

- Merged PR and commit reconciliation
- Measured report and source run
- Architecture and runbook
- Generated mappings and evidence index
- Unsigned bounded review request